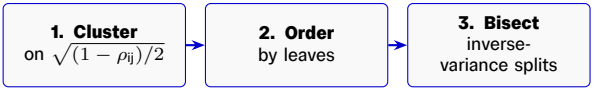
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Portfolio Optimisation with Causal Networks

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Motivation: correlation is undirected; HRP requires symmetric inputs

Hierarchical Risk Parity[1]



- Mean-variance optimisation[2] is ill-conditioned and unstable.
- HRP avoids this instability by never inverting the covariance matrix.
- But correlation can be spurious[3], and records co-movement, not causal influence.

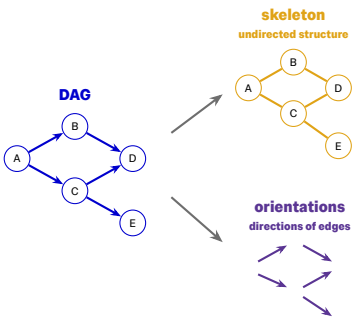
Causal discovery

- Causal discovery algorithms (NOTEARS[4], DYNOTEARS[5]) fit DAGs, giving directed causal links between assets.
- Applied to equities before[6], but not within hierarchical allocators.
- HRP's clustering requires a symmetric matrix: DAGs' **skeletons** are symmetric, but not their **edge orientations**.
- So the two components have to be input differently and their values measured separately.

[1] López de Prado (2016); [2] Markowitz (1952); [3] Simon (1954); [4] Zheng et al. (2018); [5] Pamfil et al. (2020); [6] Howard, Lohre & Mudde (2025)

Research question

When the correlation matrix in hierarchical portfolio allocation is replaced by a causal graph, is there a performance improvement, and how much of any improvement is attributable to the graph's skeleton versus its edge orientations?



- Sub-questions on the mechanisms:
  - Is the **skeleton's** effect specific to the discovered graph, or is it replicated by any sparse network of similar density?
  - Is the **edge orientations'** effect due to directionality, or to some other property of the implied covariance?
- Three challenges:
  1. Attributing performance to the graph's components (needs a credible, reproducible graph).
  2. Getting an asymmetric matrix into HRP.
  3. Inference at realistic effect sizes.

Answer: no performance improvement. The **skeleton** has a negative effect, **orientations** have a positive effect, but neither effect needs the graph.

Prior research: no direct allocation from a discovered causal graph

Hierarchical allocation and backtest overfitting

- HRP[1], HERC[7], HSP[8].
- Backtest overfitting and multiple comparisons problem: Deflated Sharpe[9], Hansen's SPA[10], Model Confidence Set[11].

Causal graphs in investing

- Howard et al.[6] apply DYNOTEARS to the S&P 500. The contemporaneous matrix is used for similarity and centrality, but not portfolio weights.

The gap

	discovery	weights	asymm.	decomp.
HRP, HSP	–	–	–	–
MST / PMFG	–	✓	–	–
Granger nets	–	✓	✓	–
Howard et al.[6]	✓	–	–	–
This thesis	✓	✓	✓	✓

Causal discovery methods

- NOTEARS/DYNOTEARS[5] fit a linear SEM with an acyclicality constraint to variables, which scales well.
- VARLiNGAM[12] identifies DAGs by assessing the independence of non-Gaussian residuals.

[7] Raffinot (2018); [8] Rodríguez Domínguez (2023); [9] Bailey & López de Prado (2014); [10] Hansen (2005); [11] Hansen et al. (2011); [12] Hyvärinen et al. (2010)

Discovering the causal graph over the assets and drivers

Data

- 99 US large-cap assets (S&P 100 proxy), CRSP via WRDS, fixed 2007–2024. Survivorship bias (excluding dropouts) affects levels, not pairwise comparisons.
- 33 plausibly exogenous macro drivers (rates, credit, FX, commodities, VIX), standardised per window.
- 4 estimation windows: 189, 252, 378, and 504 days; 215 monthly rebalances.

Causal discovery algorithms

- Primarily **DYNOTEARS**[5] (p $= 1$, $\lambda = 0.05$).
- **VARLiNGAM**[12,13] comparator; **ridge-Granger**[14] as ancillary control on edge magnitudes.

	D	A
D	drivers→drivers	drivers→assets
A	assets→drivers **forbidden**	assets→assets = M

- M: contemporaneous asset→asset block of the DAG, net of drivers' effects. Edge density: 6–7% for DYNOTEARS, 32% for VARLiNGAM at 189 days (due to noisy adaptive lasso penalties).

[13] Shimizu et al. (2011); [14] Granger (1969). Data: CRSP via WRDS, FRED, Yahoo Finance

Incorporating asymmetric edge orientations into hierarchical allocators

Structural equation model[15]

$$(\mathrm{I} - \mathrm{M}^\top)\,\mathrm{x} = \varepsilon$$

$$\mathrm{T} \equiv (\mathrm{I} - \mathrm{M}^\top)^{-1} = \textstyle\sum_{k\geq 0}(\mathrm{M}^\top)^k \quad \text{(exact, as M nilpotent)}$$

$$\Sigma_{\mathrm{SEM}} = \mathrm{D}_\sigma \left(\mathrm{T}\,\Sigma_\varepsilon\,\mathrm{T}^\top\right)\mathrm{D}_\sigma,$$

$$\Sigma_\varepsilon = \mathrm{diag}(\sigma^2_{\varepsilon,\mathrm{i}}),\ \mathrm{D}_\sigma = \mathrm{diag}(\sigma_\mathrm{i})$$

- T_{ij} of a unit shock to asset j on asset i along every path[16]; edge orientation information enters through its asymmetry.
- Σ_ε is diagonal by design to force the model to explain dependence between assets via T, not sample correlation.
- Independent shocks remove the common market factor; the mechanism controls isolate this effect.

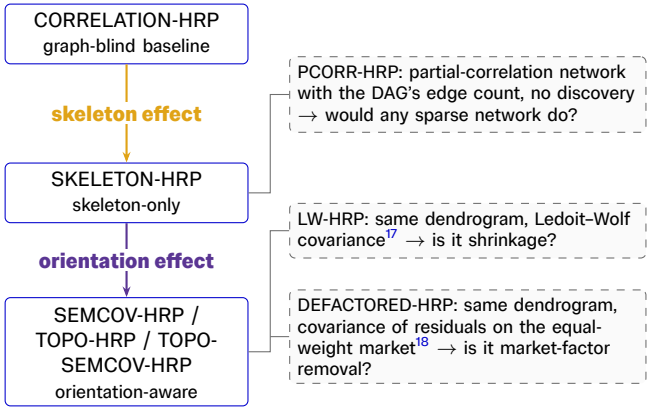
Two entry points, four allocators

ordering from	bisects on sample	bisects on Σ_{SEM}
skeleton dendrogram	SKELETON-HRP	SEMCOV-HRP
topological sort	TOPO-HRP	TOPO-SEMCOV-HRP

- Allocators are long-only, deterministic, same M.

[15] Bollen (1989); [16] Wright (1934)

Decomposing the effect: ablation design and controls to identify mechanisms



Total DAG effect = skeleton + orientation.

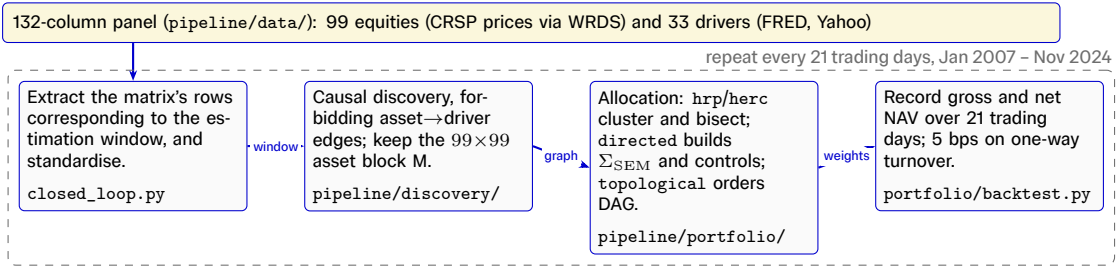
Robustness checks: UNDIRECTED-HRP; VARLiNGAM and ridge-Granger graphs; HERC; sparsity and cost sweeps.

Measurement

- Pairwise comparisons with net Sharpe, 5 bps one-way transaction costs; stationary block bootstrap[19].
- Deflated Sharpe[9,20] over all 146 configurations to account for multiplicity.
- Hansen's SPA[10,21] and MCS[11] for testing across whole groups ("families")
- Power: minimum detectable effect ≈ 0.047 (total), 0.037 (orientation).

[17] Ledoit & Wolf (2004); [18] Sharpe (1963); [19] Politis & Romano (1994); [20] Bailey & López de Prado (2012); [21] White (2000)

Implementation



Cells (discovery method, allocator, window, cost) saved in `results/<tag>/closed_loop.pkl`; launched by `scripts/run_phase_ii.py`. 227 cells total, including sweeps, Phase I, and out-of-sample testing.

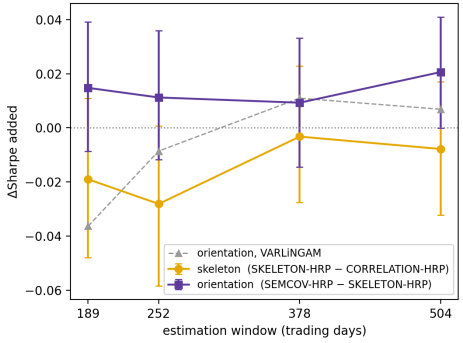
Performance metrics, comparisons, and figures in `scripts/`.

`reproduction.py` runs the complete pipeline. `causalnex`, `lingam`, and `nts-notears` are vendored.

Last-minute bugfix: All allocators accidentally had a nearest-PSD projection of its distance matrix, collapsing it to rank one. Removing it raised CORRELATION-HRP by $+0.020$ to $+0.046$ and moved the graph-fed allocators by at most 0.005, creating an apparent performance gain.

Code and collated results: github.com/j-muthu/portfolio-optimisation-causal-networks. Exact reproduction needs WRDS/CRSP access; Yahoo prices are a fallback.

Results I: no overall performance gain, skeleton negative, orientation positive



△Sharpe (p)[19]	189d	252d	378d	504d
skeleton	−0.019 (0.21)	−0.028 (0.056)	−0.003 (0.80)	−0.008 (0.56)
orientation, alloc. step	+0.015 (0.22)	+0.011 (0.35)	+0.009 (0.45)	+0.021 (0.051)
orientation, both steps	+0.016 (0.31)	+0.007 (0.61)	+0.015 (0.33)	+0.024 (0.086)
total	−0.004 (0.82)	−0.017 (0.36)	+0.006 (0.73)	+0.013 (0.44)

All contrasts on DYNOTEARS graphs.

- Only 6/20 DYNOTEARS cells beat HRP; 1/20 for VARLiNGAM.
- All 12 orientation vs. skeleton contrasts are non-negative but insignificant.
- Under VARLiNGAM graphs, the orientation contrasts show no consistent sign.
- Against the stricter skeleton allocator, the positive orientation effect falls.

Results II: neither effect needs a graph

control	tests whether. . .	189	252	378	504	reproduces it?
LW-HRP[17]	shrinkage explains the orientation gain (LW-HRP − SKELETON-HRP)	-0.002 (0.86)	-0.005 (0.78)	-0.022 (0.19)	-0.019 (0.34)	**no**
DEFACTORED-HRP[18,22]	market-factor removal explains the orientation gain (DEFACTORED-HRP − SKELETON-HRP)	$+0.019$ (0.49)	$+0.010$ (0.69)	$+0.020$ (0.42)	$+0.027$ (0.26)	**yes**
PCORR-HRP	any sparse network reproduces the skeleton cost (PCORR-HRP − SKELETON-HRP)	-0.020 (0.073)	$+0.012$ (0.23)	-0.005 (0.62)	$+0.007$ (0.48)	**yes**

△Sharpe and two-sided block-bootstrap p-values.

- Shrinkage does not explain orientation effect, as LW-HRP underperforms SEMCOV-HRP.
- Market-factor removal explains orientation effect, as DEFACTORED-HRP matches SEMCOV-HRP (Σ_{SEM}'s independent shocks remove the common market factor).
- The DAG's density and sparsity alone explain the skeleton's effect, as PCORR-HRP matches SKELETON-HRP.

- Ridge-Granger[14] keeps directions but degrades magnitudes, which weakens the allocators bisecting on Σ_{SEM} – direction alone does not drive performance.
- Market-factor removal driving orientation gain mirrors network-portfolio findings on centrality of assets[23,24].

[22] Laloux et al. (1999); [23] Pozzi et al. (2013); [24] Peralta & Zareei (2016)

Results III: nothing survives the search; out of sample results are mixed

Hansen's SPA p-values[10]

family	set	189	252	378	504	pooled
graph-fed vs baseline	full (7)	0.440	0.785	0.346	0.235	0.823
	narrowed (5)	0.673	0.886	0.273	0.213	0.827
oriented vs skeleton	full	0.217	0.299	0.328	0.173	0.451
	narrowed (3)	0.361	0.333	0.175	0.076	0.399

- Similarly, allocators' Deflated Sharpe ratios[9] never reach 5% significance.
- The 90% Model Confidence Set[11] keeps 16 of 17 strategies and drops only HSP.
- Point estimates are under 1.5× the bootstrap SEs; observed effects are about half the MDE.

Out of sample testing, Jan 2025 – Aug 2026

△Sharpe	189	252	378	504
orientation	$+0.035$	$+0.050$	$+0.105$	$+0.036$
skeleton	$+0.107$	$+0.071$	-0.009	$+0.100$
SEMCOV-HRP − DEFACTORED-HRP	-0.090	-0.001	$+0.005$	-0.001

- No statistically significant results.
- Orientation positive as in sample; the skeleton's contribution flips to positive at 3/4 windows.

Robustness checks

- Using HERC[7] as a framework instead of HRP reproduces the negative skeleton and positive orientation effects; still insignificant.
- Howard et al.[6] note causal DAGs may help in stressed regimes, but I find graph-fed allocators underperform HRP/HERC in recessions.

Conclusions: replacing correlations with DAGs does not improve performance

Contributions

1. An ablation design with graph-free mechanism controls, fed by a reproducible DAG.
2. Asymmetric inputs into hierarchical allocation: Σ_{SEM} at the allocation step, topological order at the ordering step.
3. The finding: no gain at any window; the two components are insignificant, offset each other, and are reproducible without a graph.

Implications

- Market-factor removal performs as well as causal discovery (i.e. as well as standard HRP).
- Causal allocators should be tested via ablation and mechanism controls to identify the true performance effect drivers.
- Claims about "causality" improving prediction and allocation[25] should be treated with skepticism!

Limitations

- Underpowered: effects ≤ 0.03 vs MDE 0.04–0.05.
- Orientation effect is DYNOTEARS-specific; long-only US large caps only.
- Three disclosed post hoc changes, included in the search-space correction.
- R^2-sortability[26] and λ untested.

Future work: NTS-NOTEARS[27]; risk parity over structural shocks.

The graph's skeleton is a worse clustering input than correlation, and its orientations only help through a mechanism that needs no graph.

[25] López de Prado (2023), [26] Reisach et al. (2023), [27] Sun et al. (2023)

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Thank you

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